

MCAE0406: MOBILE APPLICATION DEVELOPMENT II

Objective: To introduce students to cross-platform mobile application development using Flutter and Dart, enabling them to build modern, responsive, and feature-rich applications for Android and iOS platforms.

Credits:03

L-T-P-J: 3-0-0-0

Module No.	Content	Teaching Hours
I	Preliminaries - Introduction to Mobile Apps: Need for Mobile App Development, Examples of Mobile Applications in Different Categories. Cross-Platform vs Native Development - Overview of Mobility Concepts and Different Mobile Platforms (Android, iOS, Hybrid) Introduction to Flutter and Mobile App Development - Importance of Flutter Framework, Features and Advantages of Flutter, Flutter Architecture Overview (Dart, Widgets, Engine, Foundation Libraries), Comparison - Flutter vs React Native, Ionic, and Native Development Understanding the Dart Programming Language - Introduction to Dart, Installing Dart SDK and Configuring the Environment, Using DartPad for Hands-on Coding Practice, Dart Syntax, Writing Comments and Documentation, Conditional Statements and Operators, Functions and Object-Oriented Programming in Dart, Creating and Using Functions, Object-Oriented Programming (OOP) Concepts Flutter Project Setup and App Development Environment - Installing Flutter SDK and Configuring Path Variables, Setting up Development Environment, Setting up Emulators and Physical Devices, Project Structure in Flutter and Understanding Key Files (pubspec.yaml, lib directory), Debugging and Hot Reload/Hot Restart Flutter Widgets Overview and Basics - Introduction to Widgets (Widget Tree and Widget Lifecycle, Stateless and Stateful Widgets, Essential Widgets, Text, Image, and Container, Row and Column (Layouts), Stack and Align Widgets, Buttons, Material App and Scaffold (Customizing Scaffold (AppBar, Body, FloatingActionButton)), Exploring Assets, Fonts, and Themes for Visual Customization	19
II	Advanced Flutter Concepts, App Deployment & Industry Standards Navigation and Routing - Understanding App Structure and Navigation, Implementing Navigation Techniques - push(), pushNamed(), and Navigator.pop(), Passing Data Between Screens and Managing Routes, Named Routing and Route Arguments State Management in Flutter - Understanding State - Stateless vs Stateful Widgets, Local State Management - setState() for Small Scale Updates, Advanced State Management Techniques - Provider, Riverpod (Dependency Injection), and BLoC Pattern, Overview of GetX for State, Routes, and Controller Management. App Performance Optimization - Code Structuring and Optimizing Widget Rendering, Hot Reload vs Hot Restart, Minimizing Package Dependencies to Reduce App Size, Obfuscating Dart Code for Production Deployment Version Management and Deployment - Understanding App Versioning - Configuring pubspec.yaml, Version Naming and Build Numbers (Semantic Versioning), Packaging and Signing Flutter Apps for Android - Generating APK and AAB Builds, Keytool and Jarsigner for Code Signing, Build Commands - flutter build apk, flutter build appbundle, iOS Builds with Xcode - Configuring Flutter for iOS Deployment, Managing Certificates, Code Signing, and Provisioning Profiles. Industry Standard Tools and Practices for Mobile App Development - Code Version Control - Using Git and GitHub for Collaboration, API Integration- RESTful APIs with http package, Firebase (Firestore, Realtime Database, Cloud Storage, Firebase Authentication).	19

	Testing Flutter Applications - Debugging and Log Outputs with print() and Debug Console, Manual and Automated Testing - Unit Testing with test package , Integration Testing, Widget Testing with Flutter Test Tools, Performance Profiling and Analytics.	
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References:

- "Flutter for Beginners" by Alessandro Biessek
- "Beginning Flutter: A Hands-On Guide to App Development" by Marco L. Napoli

Outcome: After completion of course, student will be able to:

- CO1:** Understand various mobile application architectures, design principles, and development patterns specific to Flutter.
CO2: Describe the components, structure, and functionalities of the Flutter framework and the Dart programming language.
CO3: Develop cross-platform mobile applications using Flutter and its core features.
CO4: Deploy, test, and debug Flutter applications on both Android and iOS platforms

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1, PO5/PSO1
CO2	PO1, PO3/PSO1, PSO2
CO3	PO1, PO3/PSO2, PSO4
CO4	PO1/PSO1

